

COMPUTER INVESTIGATION REPORT

Hardware, Performance, Storage, Network and Security Investigation

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Investigation period: 14–18 September 2026 (from timestamps visible in supplied evidence)

Platform: Dell Vostro 260 / Ubuntu 24.04.1 LTS

Purpose: establish a documented hardware/software baseline, reproduce performance behaviour, record storage, network and startup measurements, and identify evidence-backed areas for remediation.

Evidence note. The team reports that multiple computers share the same specifications and observed issues. This revised document integrates the original report with the additional terminal screenshots supplied afterwards.

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This revised report is organised so that the new screenshots are discussed in the relevant technical sections rather than being left only as an appendix.

1. Executive Summary

The investigation establishes a baseline for the Dell Vostro 260 systems and adds direct evidence from Linux diagnostics, stress testing, SMART storage checks, startup timing, network testing and a Lynis security audit.

Key hardware: Intel Core i5-2400, approximately 4 GB DDR3 memory and a 500 GB-class Seagate SATA hard drive. Recorded CPU stress, memory stress and CPU/I/O stress tests completed successfully. CPU temperatures visible during load were approximately 51–58°C.

Storage: smartctl reports SMART overall health as **PASSED**. This is a captured health-status result, not a disk-speed benchmark.

Network: approximately **44.28 Mbit/s download**, **2.41 Mbit/s upload** and approximately **1028.6 ms latency** to the selected London server. The route evidence shows **192.168.0.69/24** via **192.168.0.1** on **enp2s0**.

Startup: systemd-analyze records **1 min 25.121 s** total startup, with the graphical target reached after approximately **1 min 17.977 s** in userspace.

Security: Lynis provides a hardening worklist covering auditing/accounting, file integrity, permissions, authentication, boot security, services, logging, network/printing settings and filesystem layout.

Important: the short stress tests did not reproduce a crash or forced shutdown. The evidence therefore supports several performance and configuration investigations, but does not demonstrate one single hardware failure.

2. Investigation Scope and Method

The original project covers baseline, CPU, RAM, storage, operating system, drivers/updates, startup, background processes, applications, security, shutdowns, cooling, physical hardware, power/battery, network, workload, reproduction, diagnostics, root cause and recommendations. The supplied evidence does not fully cover every category, so incomplete areas remain identified rather than inferred.

Evidence	Information used
Original BIOS photograph	System model, BIOS, CPU, RAM, storage and optical-drive baseline.
Original lscpu photograph	CPU architecture, topology, frequency range, cache and VT-x.
Original htop/stress photographs	CPU load, memory use, temperatures and workload completion.
Original hardware/monitor photographs	Physical components and display-cable diagnosis.
New smartctl screenshots	SMART health, drive identity and self-test history.
New speedtest/route screenshots	Throughput, latency, interface, local address and default route.
New systemd-analyze screenshot	Startup completion and graphical-target timing.
New Lynis screenshots	Security audit suggestions and hardening areas.

Evidence discrepancy to verify: the original BIOS record identifies the drive as ST3500413AS, while the later smartctl screenshot identifies ST3500418AS. The exact physical drive model should be checked on the machine.

3. Baseline and Hardware Identification

The BIOS evidence identifies the machines as Dell Vostro 260 systems and records BIOS A10 with a build date of 22 February 2013. It also records an Intel Core i5-2400, 4096 MB DDR3 at 1333 MHz and a 500.1 GB SATA hard drive.

Component	Recorded information
System	Dell Vostro 260
BIOS / firmware	A10; build date 02/22/2013
Processor	Intel Core i5-2400 @ 3.10 GHz
Cache	L2 1024 KB; L3 6144 KB
Memory	4096 MB DDR3, 1333 MHz, dual-channel (BIOS record)
Storage	500.1 GB SATA drive; exact model requires verification
Optical drive	HL-DT-ST DVD+-RW GH70N ATAPI, SATA 1
Operating system	Ubuntu 24.04.1 LTS; GNOME 46; Linux 6.14.0-35

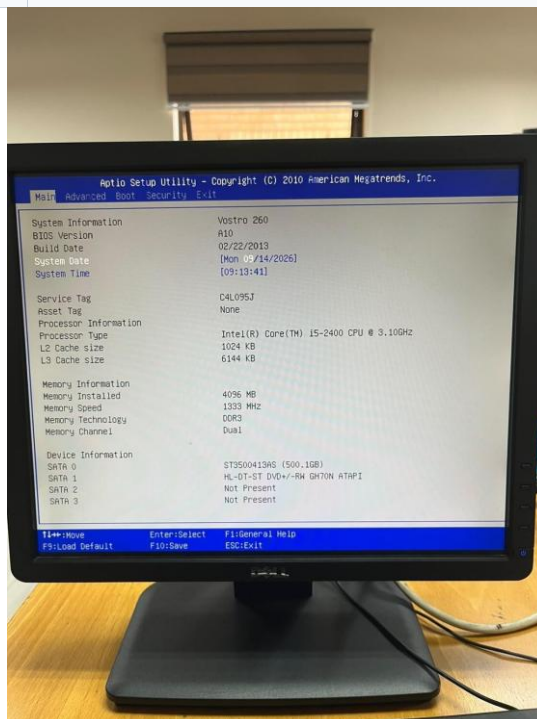


Figure 1. Original BIOS / Aptio Setup Utility evidence.

4. CPU and Memory

The lscpu evidence confirms a 64-bit x86 system using an Intel Core i5-2400. Four logical CPUs are online, matching four physical cores, with one thread per core and one socket. The reported frequency range is 1600–3400 MHz and VT-x virtualisation support is present.

CPU characteristic	Observed value
Architecture	x86_64
CPUs / cores	4 / 4
Threads per core	1
Socket	1
Frequency range	1600–3400 MHz
Virtualisation	VT-x
Cache	L1d 128 KiB; L1i 128 KiB; L2 1 MiB; L3 6 MiB

```

user04@user04-Vostro-260: ~
$ lscpu
Architecture:            x86_64
CPU op-mode(s):          32-bit, 64-bit
Address sizes:            36 bits physical, 48 bits virtual
Byte Order:               Little Endian
CPU(s):                   4
  On-line CPU(s) list:    0-3
Vendor ID:                GenuineIntel
Model name:               Intel(R) Core(TM) i5-2400 CPU @ 3.10GHz
CPU family:               6
Model:                    42
Thread(s) per core:       1
Core(s) per socket:       4
Socket(s):                 1
Stepping:                  7
CPU(s) scaling MHz:       53%
CPU max MHz:              3400.0000
CPU min MHz:              1600.0000
BogoMIPS:                  6185.53
Flags:                     fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pg
                          e mca cmov pat pse36 clflush dts acpi mmx fxsr sse
                          sse2 ht tm pbe syscall nx rdtscp lm constant_tsc ar
                          ch_perfmon pebs bts rep_good nopl xtopology nonstop
                          _tsc cpuid aperfmperf pni pclmulqdq dtes64 monitor
                          ds_cpl vmx smx est tm2 ssse3 cx16 xtpr pdcm pcid ss
                          e4_1 sse4_2 x2apic popcnt tsc_deadline_timer aes xs
                          ave avx lahf_lm epb pti ssbd ibrs ibpb stibp tpr_sh
                          adow flexpriority ept vpid xsaveopt dtherm ida arat
                          pln pts vmmi md_clear flush_l1d
Virtualization features:
  Virtualization:         VT-x
  Caches (sum of all):
    L1d:                   128 KiB (4 instances)
    L1i:                   128 KiB (4 instances)
    L2:                     1 MiB (4 instances)
    L3:                     6 MiB (1 instance)
  NUMA:
    NUMA node(s):          1
    NUMA node0 CPU(s):     0-3

```

Figure 2. Original Linux lscpu evidence.

The BIOS records 4096 MB DDR3 at 1333 MHz in dual-channel mode. htop sessions showed roughly 1.0–1.8 GB in use, with approximately 3.74 GB visible to the operating system. The original investigation associates limited memory capacity with poor multitasking.



Figure 3. Original internal hardware photograph.

5. Operating System and Background Processes

The captured environment reports Ubuntu 24.04.1 LTS, GNOME 46 and Linux kernel 6.14.0-35. The supplied evidence does not show an operating-system fault. http captures approximately 121–149 tasks and about 99 threads, including systemd, avahi, messagebus, polkitd and GNOME-related components.

The original report records that applications were difficult to test because the machines became slow or unresponsive during multitasking. Security and shutdown testing were not completed in the original investigation. The new Lynis evidence is therefore presented as a separate audit result.

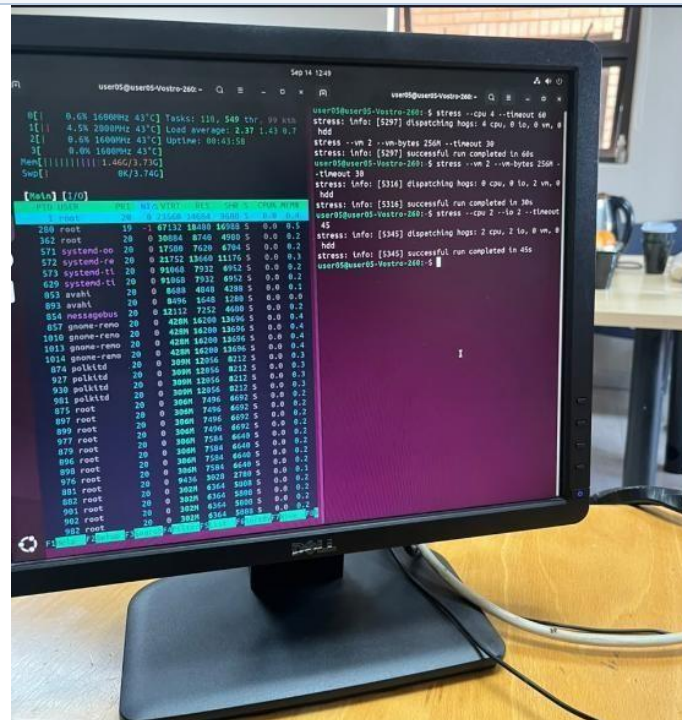


Figure 4. Original http / terminal monitoring evidence.

6. Storage and SMART Diagnostics

The new **smartctl -a /dev/sda** evidence identifies a Seagate Barracuda 7200.12, model **ST3500418AS**, approximately 500 GB, 512-byte logical/physical sectors, 7200 RPM and SATA 2.6 / 3.0 Gb/s capability. SMART overall-health self-assessment is **PASSED**.

The screenshot also shows smartmontools installation and a separate **smartctl -H /dev/sda** health check reporting PASSED. The selective self-test history contains short offline entries, including tests marked “Aborted by host” and tests marked as completed without error.

Interpretation: SMART PASSED should not be treated as a performance benchmark. The HDD can still contribute to slow startup or application response even when the captured SMART health status is PASSED.

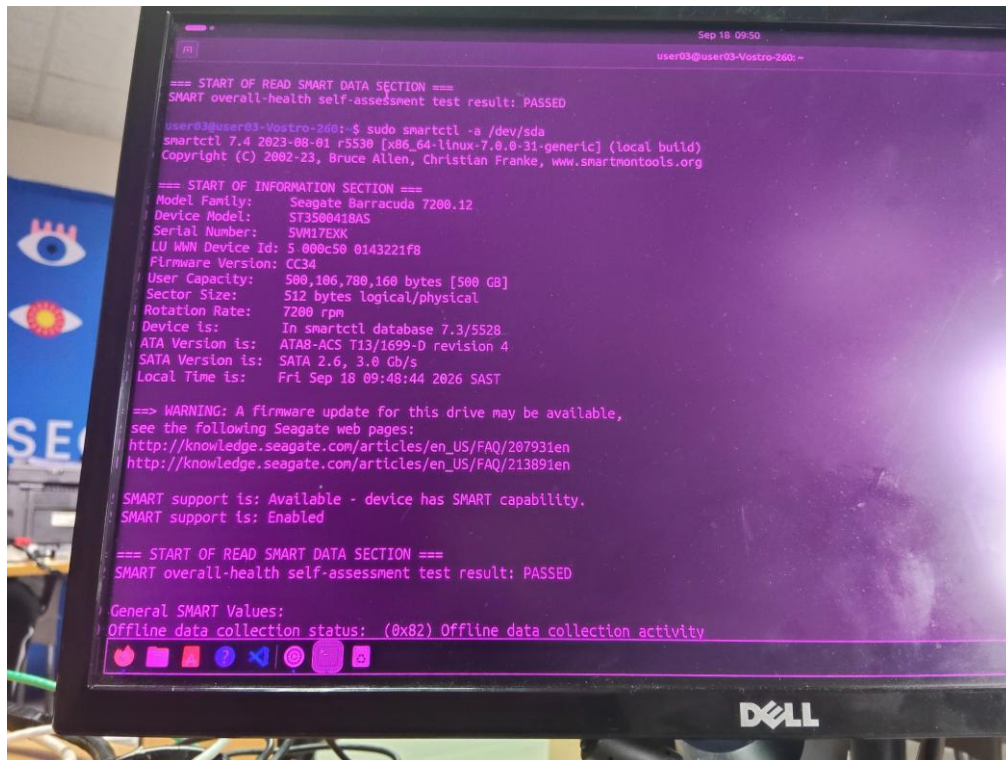


Figure 5. smartctl drive identification and SMART health result.

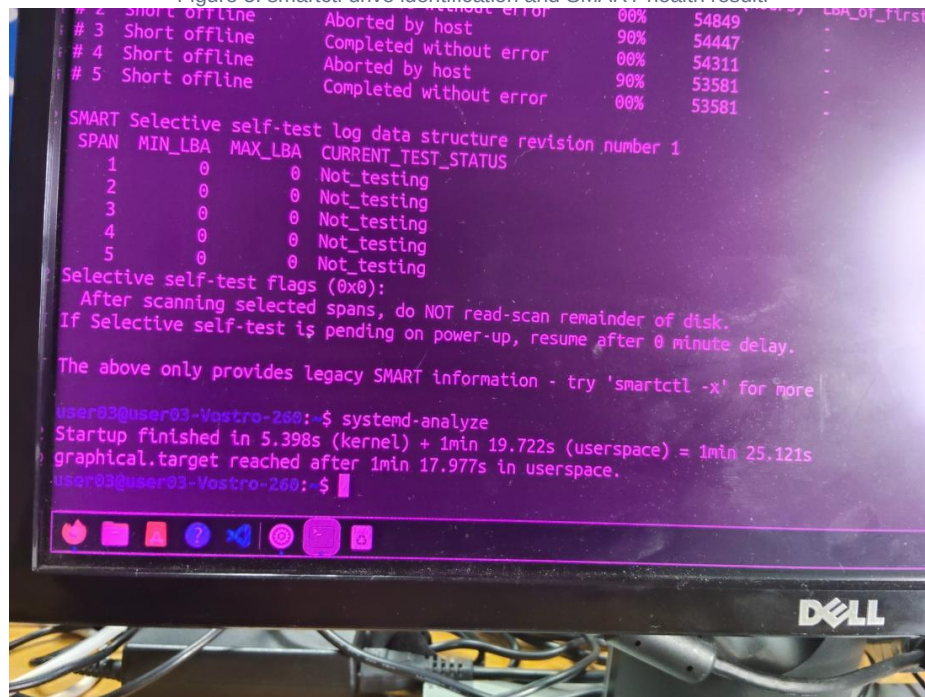


Figure 6. SMART self-test history and systemd-analyze timing.

7. Startup Performance

The supplied systemd-analyze screenshot records **5.398 s kernel + 1 min 19.722 s userspace = 1 min 25.121 s total**. The graphical target was reached after approximately **1 min 17.977 s** in userspace.

This provides a concrete measurement for the previously reported slow boot behaviour. It does not by itself identify the service responsible for the userspace delay. Follow-up testing should use **systemd-analyze blame** and **systemd-analyze critical-chain**.

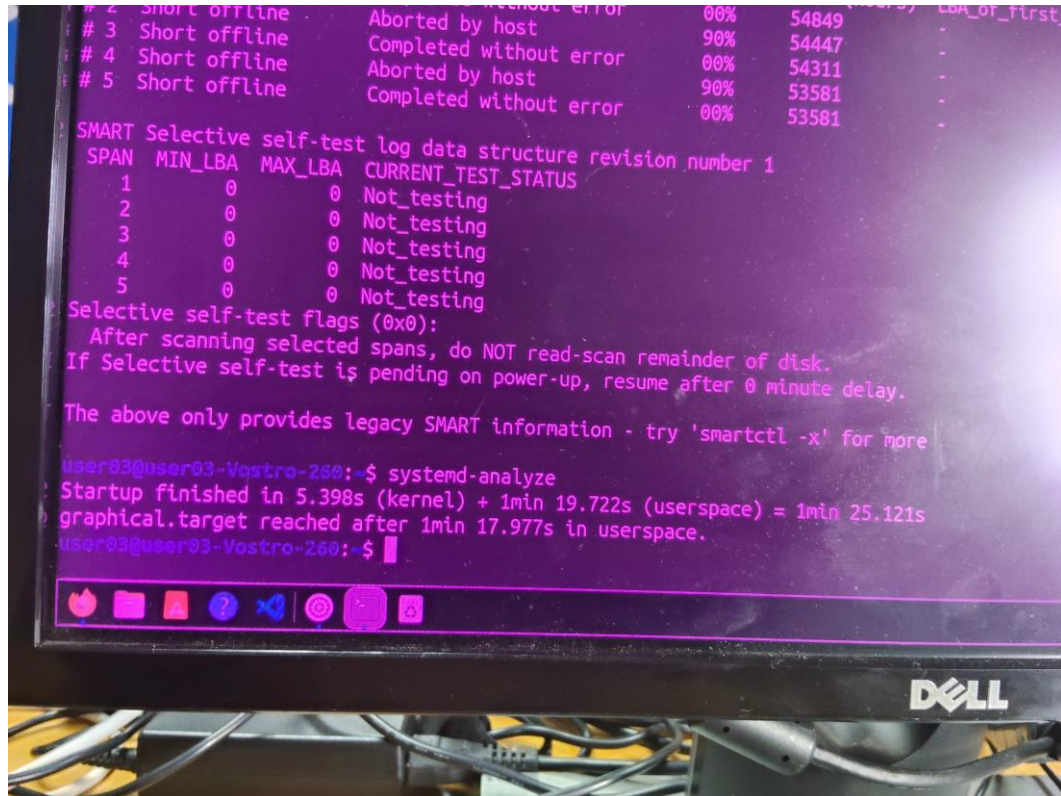


Figure 7. Captured systemd-analyze startup timing.

8. Network Investigation

The new network evidence records approximately **44.28 Mbit/s download** and **2.41 Mbit/s upload**. The selected test server was in London and the observed latency was approximately **1028.6 ms**.

The routing output shows **enp2s0**, source/local address **192.168.0.69/24** and default route **192.168.0.1**. The screenshot also contains a failed address-command attempt; the successful route output provides the interface and source address.

Measurement	Observed result
Download	44.28 Mbit/s
Upload	2.41 Mbit/s

Latency	1028.6 ms to selected London server
Interface	enp2s0
Local IPv4	192.168.0.69/24
Default gateway	192.168.0.1

Interpretation: the high observed latency is relevant to the team's report of slow network-dependent work. Because one speed test is a snapshot, repeated tests to nearby servers and at different times are needed before assigning the cause to the local network or upstream path.

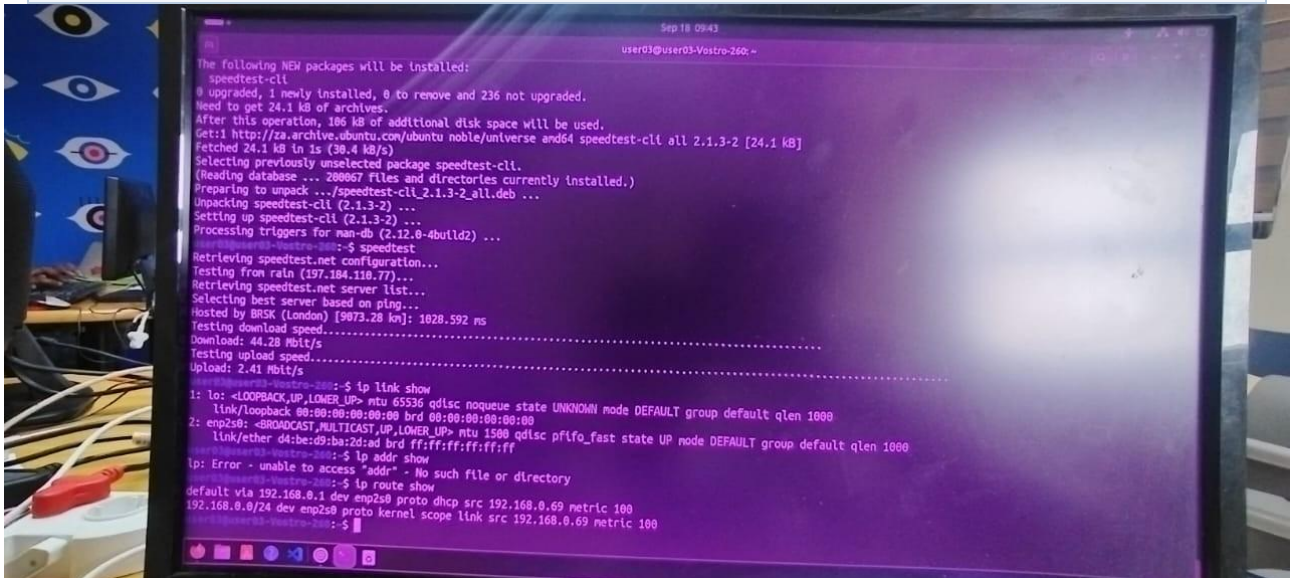


Figure 8. Speedtest CLI and Linux routing evidence.

9. Cooling and Workload Testing

During captured load sessions, CPU temperatures were approximately 51–58°C, with the highest displayed value around 58°C while utilisation was at or near full load. The original report also notes that thermal paste appeared old during physical inspection; this is a visual observation rather than a measured thermal fault.

Test	Configuration	Result
CPU stress	4 CPU workers; 60 seconds	Completed successfully
Memory stress	2 VM workers; 256 MB each; 30 seconds	Completed successfully
CPU + I/O	2 CPU + 2 I/O workers; 45 seconds	Completed successfully

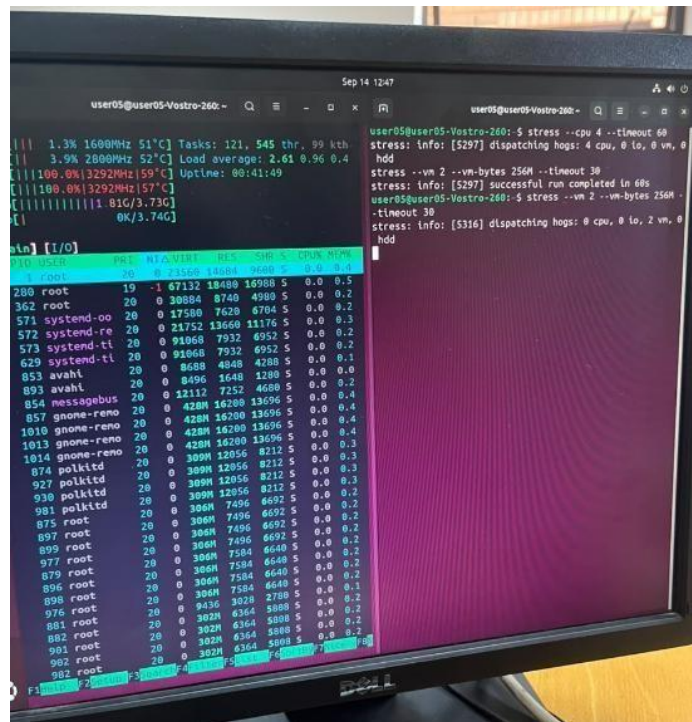


Figure 9. Original stress/htop evidence.

10. Monitor and Physical Hardware

One monitor displayed incorrect/purple colours even after factory reset. The investigation linked the problem to the VGA cable; replacing the VGA connection with HDMI restored normal colour. The supplied evidence therefore supports a cable-path diagnosis rather than a demonstrated monitor-panel failure.

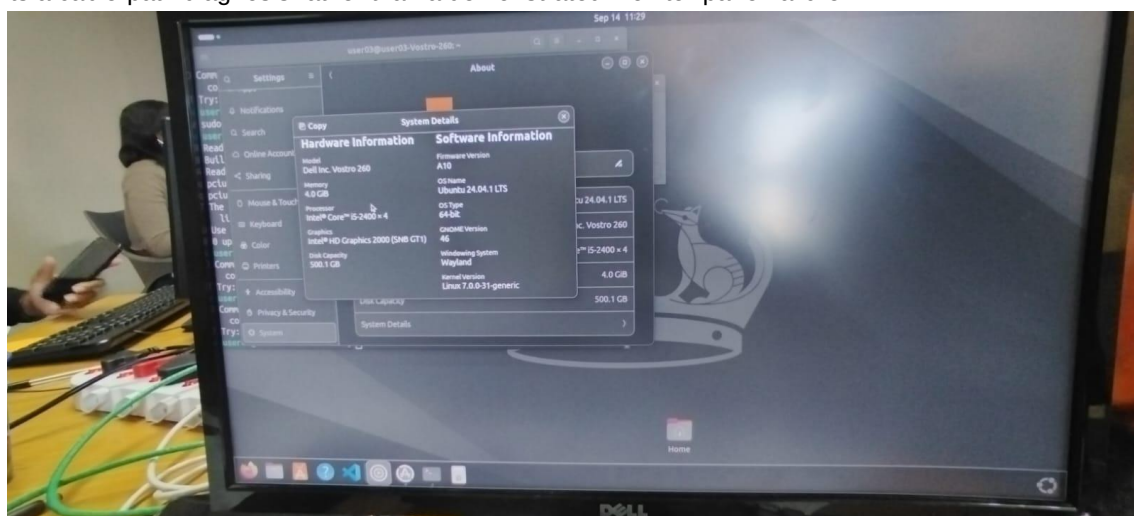


Figure 10. Original monitor evidence after the hardware change.



Figure 11. Original http/stress evidence.

The original physical inspection records two 2 GB RAM modules, a network card, battery and CPU/cooling hardware. It also notes that the thermal paste appeared old. Future RAM or cooling work should verify the actual hardware configuration before changes are made.

11. Security Audit — Lynis Findings

The new screenshots show a Lynis audit producing security-hardening recommendations. These should be treated as **audit suggestions requiring review**, not as proof that every item is an active vulnerability.

Area	Observed Lynis recommendation
Auditing/accounting	Review process accounting, sysstat/audit collection and auditd-related controls.
File integrity	Consider a file-integrity monitoring tool for critical and sensitive files.
Permissions	Review sensitive file permissions and use appropriate chmod/chown controls.
Kernel/sysctl	Review sysctl values that differ from the Lynis profile before changing them.
Root-only tools	Consider restricting compilers and similar tools to authorised users.
Malware scanning	Consider periodic filesystem scanning with an approved security tool.
Boot security	Consider protecting GRUB configuration against unauthorised modification.

Services	Review system services and harden or disable services that are not required.
Authentication	Review password hashing rounds, expiry and password-age policy.
Logging	Review external logging/archiving and deleted files that remain in use.
Access banners	Consider /etc/issue and /etc/issue.net warnings.
Network/printing	Review legacy network protocol settings and CUPS configuration.
Filesystem layout	Consider separate /home and /tmp partitions where appropriate.

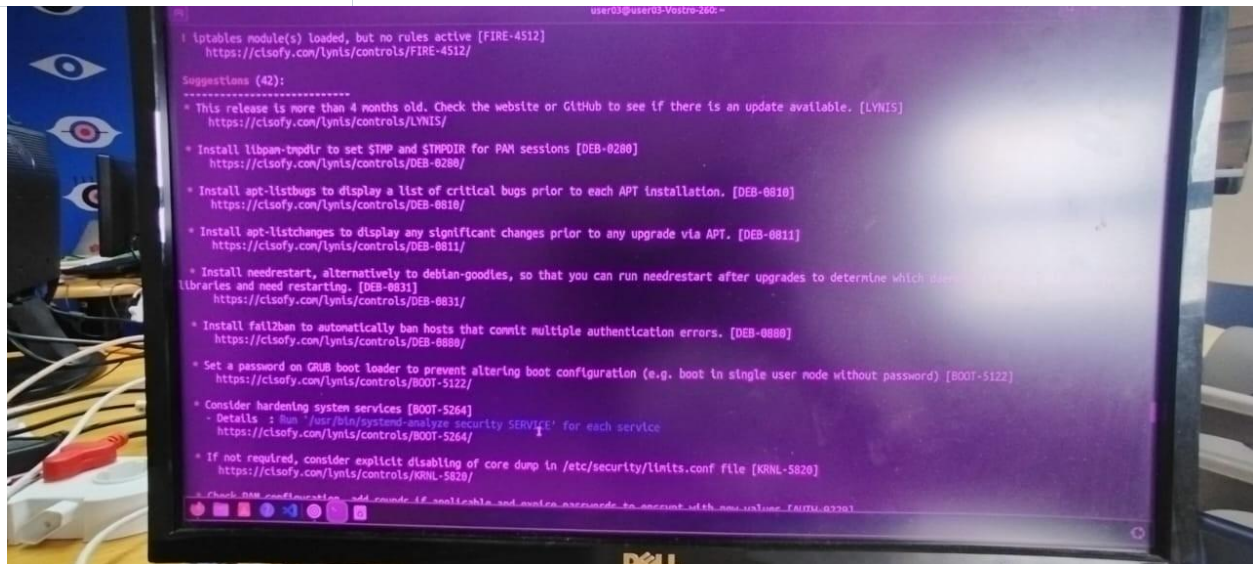


Figure 12. Lynis package, service and system-hardening suggestions.

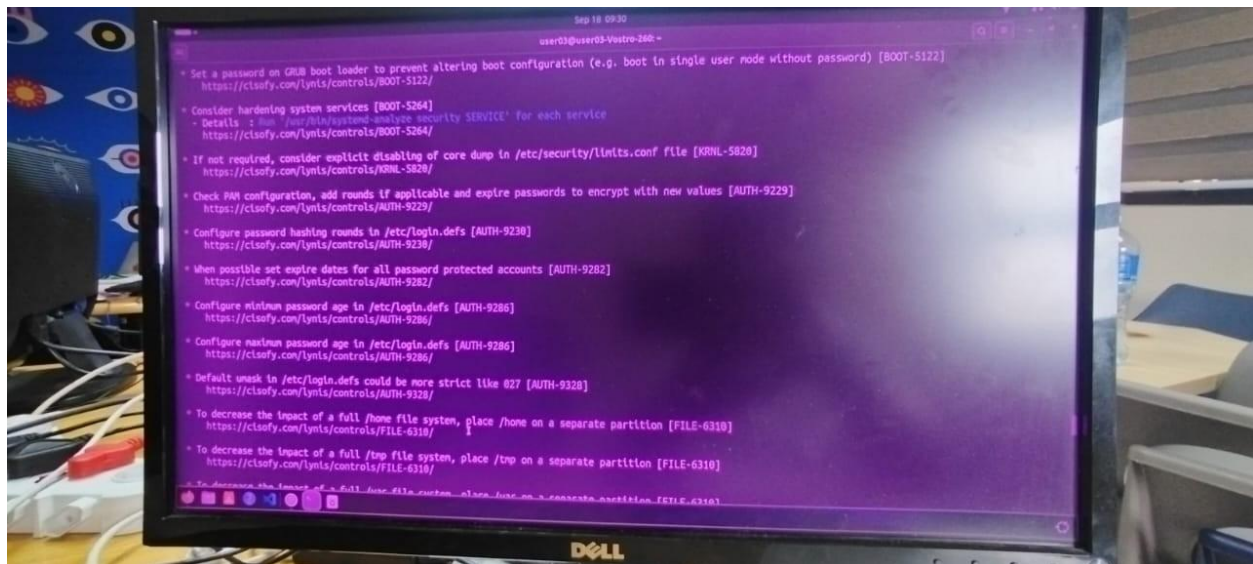


Figure 13. Lynis boot, authentication and filesystem suggestions.



Figure 14. Lynis network, printing, logging and banner suggestions.

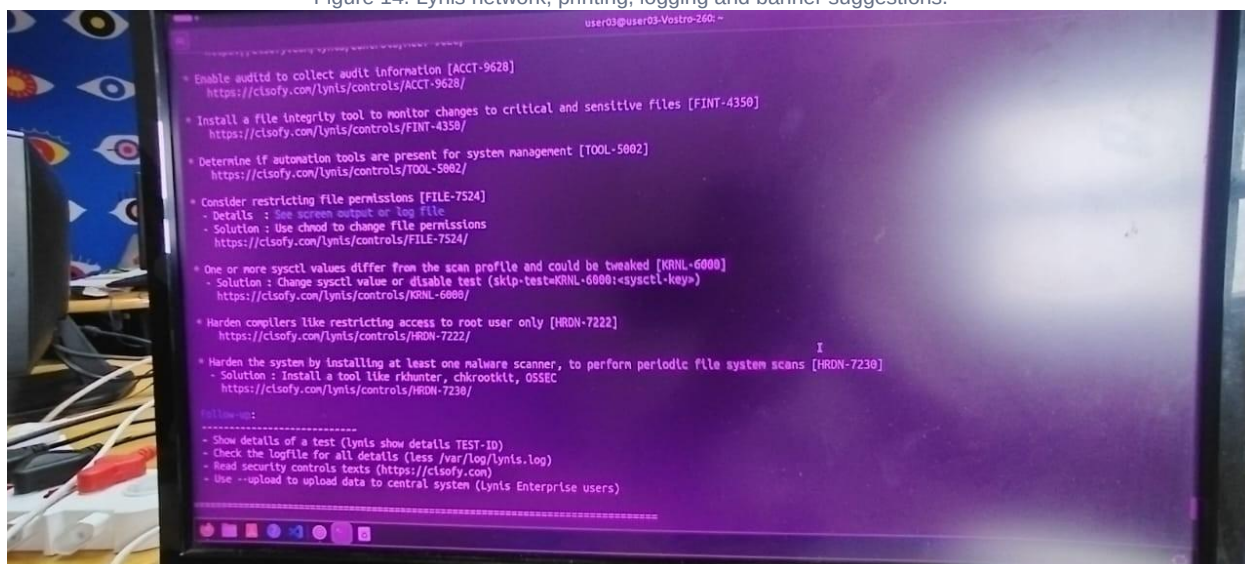


Figure 15. Lynis audit, file-integrity, permissions, kernel and malware-scanning suggestions.

The screenshots also report that the Lynis release was more than four months old at the time of the scan. For a production security review, update the audit tool and repeat the scan before finalising the hardening baseline.

12. Reproduction and Diagnostics

The stress tests provide a controlled reproduction environment for high CPU, memory and I/O demand. In the supplied sessions the system continued displaying monitoring information while the stress commands ran and the commands reported successful completion. No crash, forced shutdown or visible lock-up was captured.

The monitor fault was reproduced and then resolved by changing the cable path from VGA to HDMI. This is a direct physical-path diagnosis supported by the photographs.

Tool	Purpose
lscpu	CPU architecture, topology, frequency, cache and virtualisation.
htop	Live CPU, memory, process and temperature monitoring.
stress	Controlled CPU, memory and CPU/I/O workload generation.
smartctl	Drive identity and SMART health/self-test information.
speedtest-cli	Observed Internet throughput and latency.
ip route	Interface/source address and default gateway.
systemd-analyze	Startup timing measurement.
Lynis	Security audit and hardening recommendations.

13. Findings, Root Cause and Recommendations

The evidence points to several different issue classes rather than one demonstrated failure: slow application response/startup, a monitor colour fault, poor network responsiveness and limited memory capacity. The new evidence adds measurable startup/network results, SMART health status and a security-hardening worklist.

Area	Evidence-backed finding
Performance capacity	Approximately 4 GB RAM and an older 7200 RPM SATA HDD are associated in the original report with slow multitasking and boot behaviour.
Storage health	smartctl reports SMART overall health PASSED; this does not establish disk performance.
Startup	Measured startup is 1 min 25.121 s; graphical target after about 1 min 17.977 s in userspace.
Network	Observed 44.28 Mbit/s download, 2.41 Mbit/s upload and about 1028.6 ms latency to the selected London server.
Thermal/load behaviour	Captured load sessions show about 51–58°C and successful completion of the recorded stress tests.
Display	Purple colour issue was linked to VGA cabling; normal colour returned after HDMI was used.
Security	Lynis identifies multiple hardening areas for review; the screenshots do not establish that each item is an active vulnerability.

Recommended next steps

- 1 **Verify storage identity and performance:** confirm the exact drive model, run smartctl -x and perform an appropriate disk benchmark.
- 2 **Improve storage performance:** the original team recommendation is to replace the HDD with an SSD for faster startup and application response.
- 3 **Increase memory capacity:** verify physical slot configuration and consider additional RAM for multitasking.
- 4 **Investigate startup:** use systemd-analyze blame and critical-chain to identify contributors to the userspace delay.
- 5 **Repeat network testing:** use several nearby servers and different times; record latency, packet loss and throughput.
- 6 **Review Lynis recommendations:** update Lynis, assess each recommendation against operational requirements, document approved changes and repeat the audit.
- 7 **Cooling maintenance:** inspect the heatsink/fan and replace old thermal compound if authorised by the maintenance process.

- 8 **Complete missing tests:** application performance, security verification, shutdown behaviour, power/battery and fuller network testing remain outstanding.

14. Commands and Evidence Notes

- lscpu
- htop
- stress --cpu 4 --timeout 60
- stress --vm 2 --vm-bytes 256M --timeout 30
- stress --cpu 2 --io 2 --timeout 45
- sudo smartctl -H /dev/sda
- sudo smartctl -a /dev/sda
- systemd-analyze
- speedtest
- ip link show
- ip route show
- Lynis audit output / recommendations

Evidence limitations and verification items

- This report is based on photographs and terminal output supplied by the team; commands were not rerun during document preparation.
- The BIOS and smartctl records show different drive model numbers and this must be verified.
- The claim that all tested computers share the same specifications/issues is preserved as a team statement and was not independently verified for every machine.
- Lynis recommendations are not, by themselves, proof of exploitable vulnerabilities.
- The network result is a snapshot to a selected server and should be repeated.

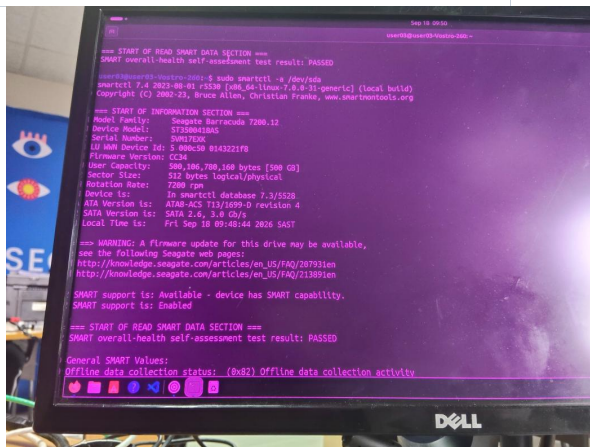
Appendix A — Integrated Screenshot Evidence

The additional screenshots are integrated into the technical sections above and indexed here for traceability.

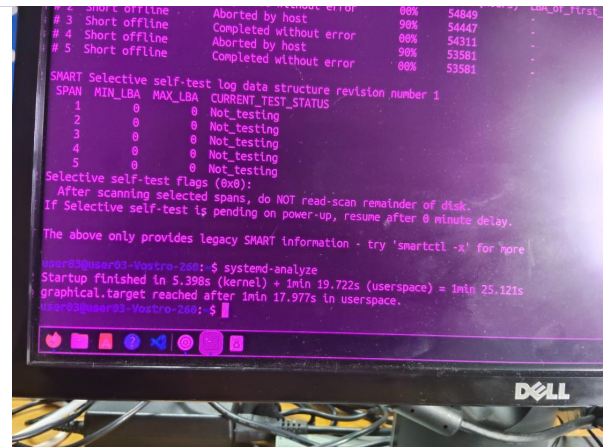
Evidence	Where used
smartctl screenshots	Storage and startup sections
speedtest / route screenshots	Network section

Lynis screenshots

Security section



Screenshot A1. SMART drive identity and health result.



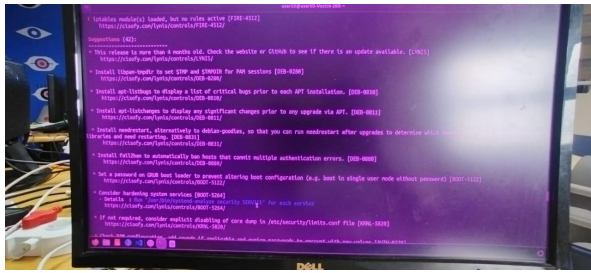
Screenshot A2. SMART self-test history and startup timing.



Screenshot A3. Speedtest and routing information.



Screenshot A4. Additional speedtest/network terminal evidence.



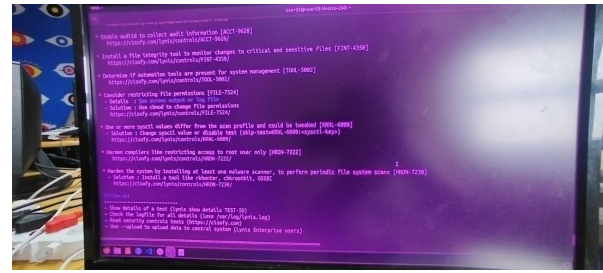
Screenshot A5. Lynis security audit suggestions.



Screenshot A6. Lynis boot, authentication and filesystem suggestions.



Screenshot A7. Lynis network, printing, logging and banner suggestions. kernel suggestions.



Screenshot A8. Lynis audit, file-integrity, permissions and suggestions.

End of Report

This revised version consolidates the original investigation narrative with the supplied screenshot evidence, improves structure and wording, and separates measured results from recommendations and items requiring verification.